

XIANGYU JIANG

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EDUCATION

Southwestern University of Finance and Economics, School of Computing and Artificial Intelligence, *Ph.D. (Admitted, 2026)* 2026

- **Status:** Admitted, 2026
- **Programme:** Artificial Intelligence
- **Research Focus:** Trustworthy generative AI, conformal risk control, and reliable machine learning

University of Sussex, School of Engineering and Informatics, *M.Sc.* Sep 2024 – Jun 2026

- **Programme:** Artificial Intelligence and Adaptive Systems
- **Core Modules:** Machine Learning, Natural Language Processing, Advanced Programming, Adaptive Systems
- **Academic Standing:** Distinction (UK highest classification)
- **Graduation Honour:** Dean's Honour Medal, Sussex Artificial Intelligence Institute
- **Research Focus:** Physics-Informed Neural Networks, Vision-Language Models, Model Compression & Quantization

Taizhou Institute of Sci.&Tech.NJUST., School of EEE, *B.Eng.* Sep 2019 – Jun 2023

- **Major:** Electrical Engineering and Intelligent Control
- **Core Modules:** Signals and Systems, Digital Signal Processing, Embedded Systems, Programming

RESEARCH INTERESTS

- Trustworthy Generative AI: retrieval-augmented generation, citation faithfulness, factuality evaluation, and risk control
- Conformal Prediction and Uncertainty Quantification: finite-sample guarantees, adaptive conformal inference, and robustness under distribution shift
- Reliable Machine Learning with Generative Priors: diffusion-model priors, score-based modeling, and calibration-aware learning
- Edge Intelligence and IoT Foundation Models: heterogeneous sensor representation learning, cross-modal masked modeling, and collaborative sensing
- Efficient Deep Learning Systems: parameter-efficient adaptation, low-bit fine-tuning, federated learning, and on-device deployment

PUBLICATIONS

[C1] X. Jiang. "CiteGuard: Conformal False-Discovery Control for Faithful Retrieval-Augmented Generation." *ICML 2026 (CCF A, Top-tier ML Conference)*. [Accepted, Solo Author]

[C2] X. Jiang. "Score-Based Diffusion Priors for Adaptive Conformal Inference under Distribution Shift." *UAI 2026 (CCF B, Flagship Conference on Uncertainty in AI)*. [Accepted, Solo Author]

[C3] X. Jiang, et al. "EdgeSense FM: Foundation Model Driven Collaborative Sensing for IoT and Sensor Networks." *IEEE ICC 2026 (CCF C, Flagship Conference in Communications)*. [Accepted]

[C4] X. Jiang, et al. "Complexity-Guided Neural Denoising for Real-Time OFDM Communications." *IJCNN 2026 (CCF C)*. [Accepted]

[C5] X. Jiang, et al. "EdgeLM-RAN: Federated Pre-trained Sequence Models for 6G Mobile Multimedia Resource Orchestration." *IEEE WoWMoM 2026 (CCF C)*. [Accepted]

[C6] X. Jiang, et al. "CAGE: Counterfactual Attribution-Guided Decoding for Grounded Vision-Language Generation." *ICICML 2025*. [Accepted]

[C7] X. Jiang, et al. "Energy Consumption Optimization of Edge Computing Based on Reinforcement Learning." *International Conference, 2021*. [Published]

RESEARCH EXPERIENCE

Trustworthy RAG with Faithful Citation Control

Oct 2025 – Jan 2026

- Proposed CiteGuard, casting claim-level citation faithfulness in RAG answers as a multiple-testing problem
- Converted claim–evidence matching scores into conformal p-values and applied BH/BY procedures for false-discovery-rate control
- Reduced the false-discovery rate from 28–31% to below 10% while retaining 86–92% of supported claims on FEVER and Natural Questions

AI-Generated Video Detection via a Frozen World-Model Backbone

Feb 2026 – May 2026

- Proposed Arrowkit, reusing a frozen video world-model backbone to extract spatio-temporal prediction residuals as discriminative features for lightweight, drift-bounded AI-generated video detection
- Trained only a 1.25M-parameter detection head while keeping the 14.5M world-model backbone frozen, with about 47ms per-clip inference latency suitable for edge deployment
- Reached 0.9844 average (0.9550 worst-case) leave-one-out cross-domain AUROC, bounding AUROC drift to 0.03, 0.15, and 0.31 on unseen CelebV, MSR-VTT, and NPR distributions

Spiking World Model with an Explicit Transport Operator for Event Streams

Mar 2026 – Jun 2026

- Proposed Adv-SSWM, modeling motion as an explicit internal operator in a spiking state-space world model via an advection–reaction dynamics transport bottleneck
- Used a correlation velocity head to estimate a displacement field for semi-Lagrangian advection of the latent state, with an evidence-gated retention mechanism preserving ballistic transport when motion is measurable and decaying unsupported velocity below the noise floor
- Extrapolated from training speeds 1–2 to speeds 3–4 on a moving-dot OOD benchmark, reaching 0.19–0.23 top-k recall versus 0.02–0.06 for ConvLSTM-style baselines

Adaptive Conformal Inference under Distribution Shift

Mar 2026 – May 2026

- Proposed DiffConf, building distribution-shift-aware nonconformity scores from the data-manifold structure learned by a score-based diffusion model
- Combined the diffusion score field, multi-noise-scale density contrast terms, and an online recalibration mechanism that jointly updates the score function and quantile threshold
- Maintained near-target coverage on synthetic, tabular-regression, and image datasets while reducing average prediction-set size by 16–33%

Calibrated Guidance for Diffusion Policies in Offline RL

Dec 2025 – Feb 2026

- Proposed CGD, guiding diffusion policies with a calibrated lower-confidence critic instead of raw point estimates to curb out-of-distribution drift from miscalibrated critics in offline RL
- Combined an offline-trained critic ensemble, distribution-free conformal calibration, and a state-adaptive guidance schedule that turns ensemble disagreement into finite-sample uncertainty bands
- Achieved 90.3 average normalized score on D4RL locomotion and 87.8 on AntMaze navigation, reducing the OOD action rate from 12.8% to 4.3%

Foundation-Model-Driven Collaborative Sensing for IoT

Jul 2025 – Oct 2025

- Proposed EdgeSense-FM, unifying heterogeneous sensor streams into sensor tokens and learning shared representations via cross-modal masked modeling
- Introduced a diffusion-style semantic completion module for missing sensors and packet loss, with 4-bit QLoRA for federated on-device adaptation
- Improved accuracy by 2–6% across activity recognition, environmental sensing, and robustness scenarios, cutting uplink communication by over 100× via adapter-only federated updates

Federated Pre-trained Sequence Models for 6G Mobile Multimedia

Jan 2026 – Mar 2026

- Proposed EdgeLM-RAN, framing 6G multimedia resource orchestration as sequence modeling from network tokens to control policies

- Deployed lightweight pre-trained sequence models at base stations with LoRA adaptation, federated learning, and 8-bit controller compression
- Achieved 27.9% QoE improvement over proportional-fair scheduling and 16.0% over DRL baselines across 19 base stations, reducing federated communication by 490×

Complexity-Guided Neural Denoising for OFDM Communications

Oct 2025 – Jan 2026

- Turned permutation entropy, sample entropy, Lempel–Ziv complexity, and Higuchi fractal dimension into differentiable complexity priors
- Designed feature-level early fusion, a complexity-consistency regularizer, and a complexity-aware curriculum sampler for CNN, Transformer, and GNN denoisers
- Achieved 28–42% BER reduction under AWGN, colored, and impulsive noise, with 15–26% BER improvement in real SDR tests

TECHNICAL SKILLS

- **Programming:** Python, SQL; proficient in Linux environments
- **Deep Learning:** PyTorch, TensorFlow; scikit-learn, OpenCV
- **Model Training:** LoRA/QLoRA fine-tuning, DPO/ORPO alignment, TRL/PEFT, instruction data curation
- **Model Deployment:** vLLM/TensorRT-LLM, INT4/INT8 quantization, Flash-Attention, KV Cache optimization
- **Retrieval Systems:** FAISS/HNSW vector indexing, LangChain, RAG pipeline development

LANGUAGES

- **English:** CET-4/CET-6; received full English-medium instruction during master's programme; proficient in academic reading and writing
- **Chinese:** Native speaker